

Lecture 02

Probabilistic Modeling in Hydrology-Part I (Moments & Probabilistic Models)

Review on Lecture 01

❖ Introduction to the Course: Statistical Hydrology

◆ Hydrology & Statistics

◆ 수문학에서의 불확실성 분류 방법 (Handbook of Hydrology, 17.1.1)

- Inherent randomness of the driving variables (predominantly **precipitation**) and the randomness of the hydrologic system (including topographic, aquifer, and soil characteristics)
- Sampling error – the measurements that hydrologists have to work with are only a small sample from a large or potentially infinite population
- Uncertainty resulting from incorrect understanding of the processes involved (System error- errors in estimating or predicting system outputs from system inputs)

◆ Introduction to Common Representative Values

- Central Tendency, Dispersion, Asymmetry, Correlation

Today's Topic

- ❖ 이전 강의에서 봤듯이, 대표값들(평균, 표준편차, 왜곡도 등)로만 어느 불확실한 변수를 설명하기에는 어려움이 있음.
- ❖ 따라서, 대부분의 확률변수들이 주로 따르는 이미 알려진 확률분포들에 대해서 학습하고, 이를 사용해서 확률변수를 설명하고자 함.
- ❖ Probabilistic Modeling in Hydrology (Parts I & II)
 - ◆ Introduction to probability & probability distributions
 - What are Discrete/Continuous probability distributions?
 - ◆ What are the factors that determine the specific shape of the probability distributions? Parameters (매개변수)!
 - ◆ How do we decide on the "adequate" values of parameters? – P2

Review on Basic Probability Concepts

❖ Populations vs Samples (모집단 vs 표본집단)

- ◆ Parameter vs Statistic (모수 μ vs 통계치 $\hat{\mu}$ $\bar{X}$) => Probabilistic Model

❖ Random Variable (RV) (확률변수) (X, Y, Z 등 영어 대문자로 표현)

(=Stochastic Variable, 추계학적 변수)

- ◆ A variable whose values are numerical outcomes of random process (** Random/stochastic process is a complex confluence of physical variables, often too difficult to track)

◆ Few Examples of RVs

- Coin Toss (Single Trial): {0, 1} –Bernoulli (베르누이 분포)
[trajectory of coin affected by flip strength, air density, moisture, ...]
- Coin Toss (N Trials): {0,1,...,N} –Binomial (이항 분포) [N times above]

Review on Basic Probability Concepts

❖ Mutually Exclusive (배반) $P(A \cap B) = 0$

❖ Independence (독립) $P(A \cap B) = P(A) \times P(B)$

❖ Conditional Probability (조건부확률)

◆ Probability of occurrence of event A given that event B has already occurred is defined as: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

◆ In case of Independence, $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \times P(B)}{P(B)} = P(A)$

❖ Bayes Theorem (베이즈 정리)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \times P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

❖ Total Probability Theorem (전체 확률의 법칙)

◆ 대신, B_n 은 서로 Mutually Exclusive

$$P(A) = \sum_n P(A|B_n)P(B_n)$$

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- ❖ 정의: 사전확률을 통해 사후확률을 업데이트하는 방법
특히 관찰을 통해 획득한 새로운 정보로 사후 확률을 업데이트
- ❖ 대부분의 Machine Learning 방법들도 가장 먼저 가설을 설정한 후, 데이터 (=새로운 관찰 결과)를 통해 처음 설정한 가설을 업데이트하기 때문에 현재까지도 매우 유용하게 사용되는 정리
- ❖ 용어 정리
 - ◆ $P(A|B)$: A의 사후확률 (Posterior Probability)
 - ◆ $P(B|A)$: 우도 (Likelihood Probability)
 - ◆ $P(A), P(B)$: A 혹은 B의 사전 확률 (Prior Probability)

Total Probability Theorem Example

EXAMPLE 2.30 An individual has 3 different email accounts. Most of her messages, in fact 70%, come into account #1, whereas 20% come into account #2 and the remaining 10% into account #3. Of the messages into account #1, only 1% are spam, whereas the corresponding percentages for accounts #2 and #3 are 2% and 5%, respectively. What is the probability that a randomly selected message is spam?

To answer this question, let's first establish some notation:

$A_i = \{\text{message is from account \# } i\}$ for $i = 1, 2, 3$, $B = \{\text{message is spam}\}$

Then the given percentages imply that

$$P(A_1) = .70, P(A_2) = .20, P(A_3) = .10$$
$$P(B|A_1) = .01, P(B|A_2) = .02, P(B|A_3) = .05$$

Now it is simply a matter of substituting into the equation for the law of total probability:

$$P(B) = (.01)(.70) + (.02)(.20) + (.05)(.10) = .016$$

In the long run, 1.6% of this individual's messages will be spam. ■

Bayes' Theorem Example

EXAMPLE 2.31 *Incidence of a rare disease.* Only 1 in 1000 adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas an individual without the disease will show a positive test result only 2% of the time (the *sensitivity* of this test is 99% and the *specificity* is 98%; in contrast, the [Sept. 22, 2012 issue of *The Lancet*](#) reports that the first at-home HIV test has a sensitivity of only 92% and a specificity of 99.98%). If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?

To use Bayes' theorem, let A_1 = individual has the disease, A_2 = individual does not have the disease, and B = positive test result. Then $P(A_1) = .001$, $P(A_2) = .999$, $P(B|A_1) = .99$, and $P(B|A_2) = .02$. The tree diagram for this problem is in Figure 2.12.

Bayes' Theorem Example

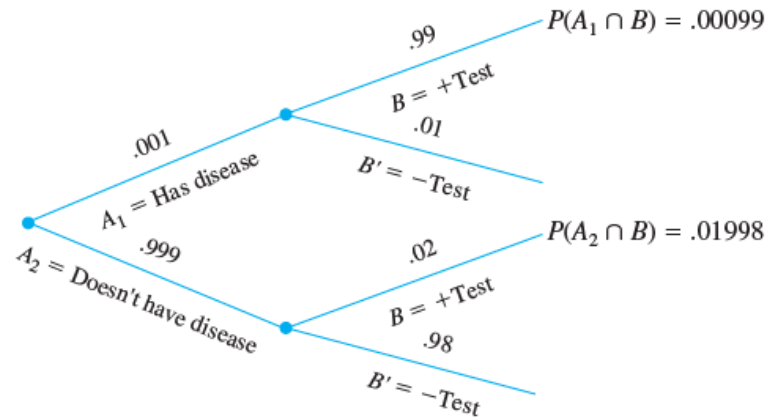


Figure 2.12 Tree diagram for the rare-disease problem

Next to each branch corresponding to a positive test result, the multiplication rule yields the recorded probabilities. Therefore, $P(B) = .00099 + .01998 = .02097$, from which we have

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{.00099}{.02097} = .047$$

This result seems counterintuitive; the diagnostic test appears so accurate that we expect someone with a positive test result to be highly likely to have the disease, whereas the computed conditional probability is only .047. However, the rarity of the disease implies that most positive test results arise from errors rather than from diseased individuals. The probability of having the disease has increased by a multiplicative factor of 47 (from prior .001 to posterior .047); but to get a further increase in the posterior probability, a diagnostic test with much smaller error rates is needed. ■

Types of Random Variables

❖ Discrete Random Variable (이산확률변수)

- ◆ Probability Mass Function (PMF) $p_X(x) = \Pr[X = x]$ $\sum p_X(x) = 1$
- ◆ Cumulative Distribution Function (CDF) $F_X(x) = \Pr[X \leq x]$
- ◆ Example Distributions: Bernoulli, Poisson, Geometric, Negative, ...

❖ Continuous Random Variable (연속확률변수)

- ◆ Probability Density Function (PDF)

$$\Pr[x_1 \leq X \leq x_2] = \int_{x_1}^{x_2} f_X(x) dx \quad \int_{-\infty}^{+\infty} f_X(x) dx = 1$$

- ◆ Cumulative Distribution Function (CDF)

$$F_X(x) = \int_{-\infty}^x f_X(z) dz \quad f_X(x) = \frac{dF_X(x)}{dx}$$

- ◆ Example Distributions:

Uniform, Normal, Log-normal, Generalized Extreme Value (GEV)

Types of Random Variables

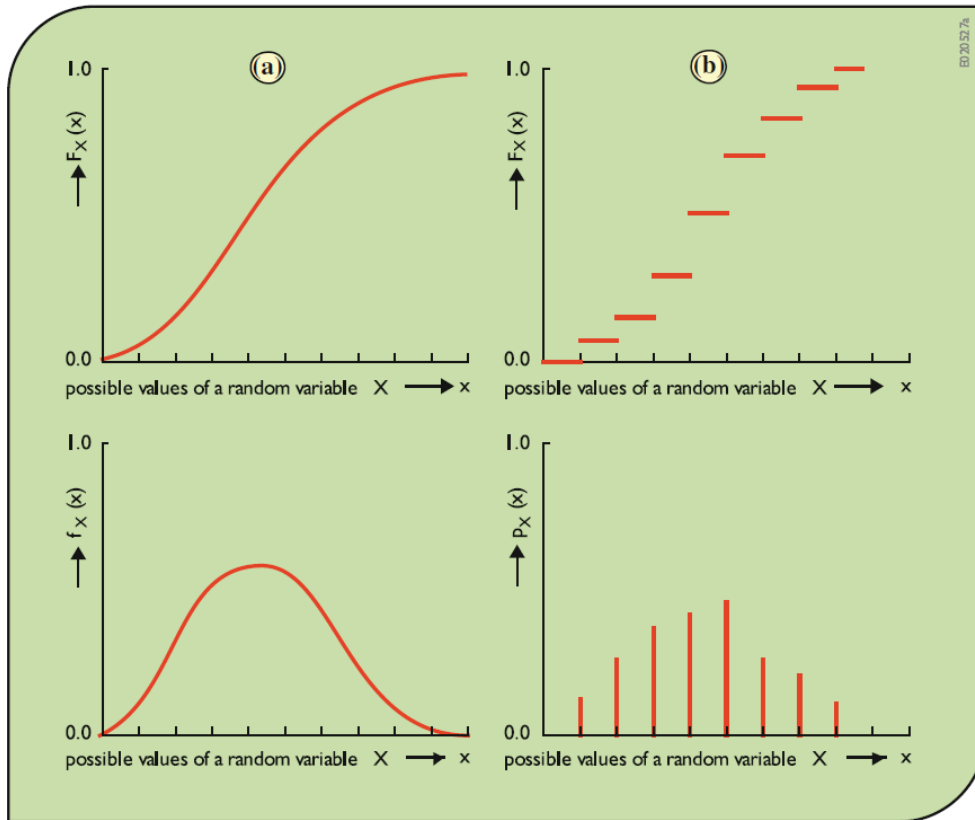


Fig. 6.1 Cumulative distribution and probability density or mass functions of random variables: **a** continuous distributions; **b** discrete distributions

*Discrete RV

$$\mu_X = E(X) = \sum_i x_i p_X(x_i)$$

$$\sigma_X^2 = Var[X] = E[(X - E[X])^2] \\ = E[X^2] - (E[X])^2$$

*Continuous RV

$$\mu_X = E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx$$

Expected Value of a Linear Function

The $h(X)$ function of interest is quite frequently a linear function $aX + b$. In this case, $E[h(X)]$ is easily computed from $E(X)$ without the need for additional summation.

PROPOSITION

$$E(aX + b) = a \cdot E(X) + b$$

(Or, using alternative notation, $\mu_{aX+b} = a \cdot \mu_X + b$)

To paraphrase, the expected value of a linear function equals the linear function evaluated at the expected value $E(X)$. Since $h(X)$ in Example 3.23 is linear and $E(X) = 2$, $E[h(X)] = 800(2) - 900 = \700 , as before.

Proof

$$\begin{aligned} E(aX + b) &= \sum_D (ax + b) \cdot p(x) = a \sum_D x \cdot p(x) + b \sum_D p(x) \\ &= aE(X) + b \end{aligned}$$

DEFINITION

Let X have pmf $p(x)$ and expected value μ . Then the **variance** of X , denoted by $V(X)$ or σ_X^2 , or just σ^2 , is

$$V(X) = \sum_D (x - \mu)^2 \cdot p(x) = E[(X - \mu)^2]$$

The **standard deviation** (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

PROPOSITION

$$V(X) = \sigma^2 = \left[\sum_D x^2 \cdot p(x) \right] - \mu^2 = E(X^2) - [E(X)]^2$$

Bernoulli Processes & Binomial Distribution

$$p_X(x) = \Pr[X = x; p] = p^x(1 - p)^{1-x}$$

- ◆ Only two possible outcomes are possible: success or failure
- ◆ The probability of occurrence of a success (or a failure) is constant
- ◆ The trials are independent (i.e., the outcome of a trial does not depend on the outcome of another trial)

$$E[X] = p$$

$$\text{Var}[X] = p$$

❖ Extension to the Binomial Distribution (0-1항분포)

- ◆ n times of Bernoulli trials

$$p_X(x) = \Pr[X = x; n, p] = \binom{n}{x} p^x(1 - p)^{n-x}$$

$$E[X] = np$$

$$\text{Var}[X] = np(1 - p) = npq$$

❖ Example 4.4

On the average, how many times will a 10-year flood occur in a 40-year flood? What is the probability that exactly this number of 10-year floods will occur in a 40-year period?

❖ Example 4.7

In order to be 90% sure that a design storm is not exceeded in a 10-year period, what should be the return period of the design storm?

Poisson Process

❖ Poisson Distribution

◆ 베르누이 과정에서 $p \rightarrow 0, n \rightarrow \infty$ 이 성립할 경우, $\lambda = np$

$$p_X(x) = \Pr[X = x; \lambda] = \frac{\lambda^x e^{-\lambda}}{x!}$$

$E[X]$

$Var[X]$

◆ Assignment! (Due Before the Class of 9/20)

이항분포의 확률 밀도 함수 (PMF)로부터 포아송 분포의 확률분포함수 (PMF)를 유도하라.

Geometric & Negative Binomial Distributions

❖ Geometric Distribution (기하 분포)

◆ 베르누이 분포에서

- 처음 성공까지 시도한 횟수 X 의 분포 $p_X(x) = p(1 - p)^{x-1}$
- 처음 성공까지 실패한 횟수 $Y=X-1$ 의 분포 $p_Y(y) = p(1 - p)^y$

$$E[X] = 1/p$$

$$Var[X] = (1 - p)/p^2$$

❖ Negative Binomial Distribution (음이항분포)

- ◆ 기하분포의 일반식으로, r 번째 성공까지 시도한 횟수 X 의 분포
(probability of the r th success occurring at the x th trial)

$$p_X(x) = \Pr[X = x; r, p] = \binom{x-1}{r-1} p^r (1 - p)^{x-r} \quad x = r, r + 1, \dots$$

$$E[X] = r * 1/p$$

$$Var[X] = r * (1 - p)/p^2$$

Summary of Discrete Distributions

Table 4.1.7 Summary of discrete distributions

Distribution	pmf, $p_X(X = x)^a$	Mean, $E[X]$	Variance, $\text{Var}[X]$	mgf
Bernoulli	$p^x(1-p)^{1-x}$, for $x = 0, 1; 0 \leq p \leq 1$	p	$p(1-p)$	$(1-p) + pe^t$
Binomial	$\binom{n}{x} p^x(1-p)^{n-x}$, for $x = 0, 1, 2, \dots, n; 0 \leq p \leq 1$	np	$np(1-p)$	$[pe^t(1-p)]^n$
Poisson	$\frac{v^x e^{-v}}{x!}$, for $x = 0, 1, 2, \dots, n; v \geq 0$	v	v	$\exp[v(e^t - 1)]$
Geometric	$p(1-p)^{x-1}$, for $x = 1, 2, 3, \dots, n; 0 \leq p \leq 1$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Negative binomial	$\binom{x-1}{r-1} p^r(1-p)^{x-r}$, for $r = 0, 1, 2, \dots, n$; $x = r, r+1, r+2, \dots; 0 \leq p \leq 1$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Alternative negative binomial	$\binom{r+y-1}{y} p^r(1-p)^y$, for $y = x - r$; for $y = 0, 1, 2, \dots$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	$\left[\frac{p}{1-(1-p)e^t} \right]^r$
Log-series	$\frac{(1-p)^x}{-x \ln(p)}$, for $x = 1, 2, 3, \dots, n; 0 \leq p \leq 1$	$-\frac{1}{\ln(p)} \frac{1-p}{p}$	$\frac{E[X]}{p} - \{E[X]\}^2$	$\frac{\ln(1-pe^t)}{\ln(1-p)}$
Multinomial	$\left(\frac{n!}{x_1! x_2! \dots x_r!} \right) p_1^{x_1} p_2^{x_2} \dots p_r^{x_r}$	See binomial for marginal distributions	See binomial for marginal distributions	Not useful
Hypergeometric	$\frac{\binom{mp}{x} \binom{m(1-p)}{n-x}}{\binom{m}{n}}$, for $x = 0, 1, 2, \min(mp, n)$	np	$np(1-p) \left(\frac{m-n}{m-1} \right)$	Not useful; see Johnson et al. (1992)

^a Change to Y variable for alternative negative binomial; some authors reverse the definitions for negative binomial and its alternative form.

Example Problem: Poisson

EXAMPLE 3.39 If a publisher of nontechnical books takes great pains to ensure that its books are free of typographical errors, so that the probability of any given page containing at least one such error is .005 and errors are independent from page to page, what is the probability that one of its 600-page novels will contain exactly one page with errors? At most three pages with errors?

With S denoting a page containing at least one error and F an error-free page, the number X of pages containing at least one error is a binomial rv with $n = 600$ and $p = .005$, so $np = 3$. We wish

$$P(X = 1) = b(1; 600, .005) \approx p(1; 3) = \frac{e^{-3}(3)^1}{1!} = .14936$$

The binomial value is $b(1; 600, .005) = .14899$, so the approximation is very good.

Similarly,

$$P(X \leq 3) \approx \sum_{x=0}^3 p(x; 3) = F(3; 3) = .647$$

which to three-decimal-place accuracy is identical to $B(3; 600, .005)$. ■

Part II:

Continuous Distributions

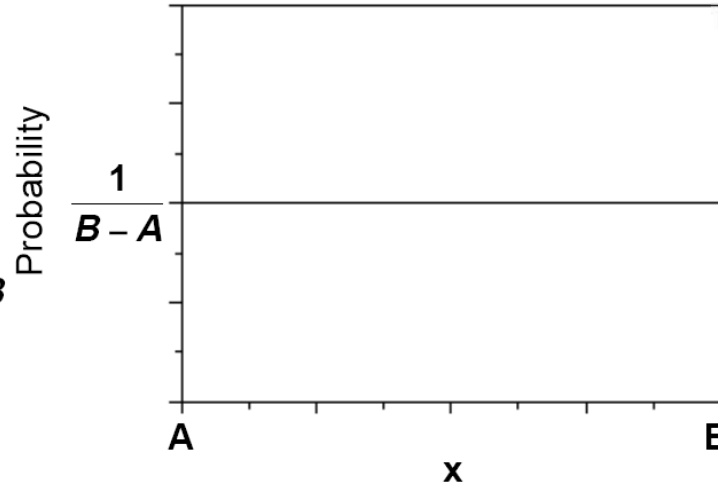
Uniform & Normal Distributions

Uniform distribution

$$f(x) = \frac{1}{B-A} \text{ for } A \leq x \leq B$$

$$F(x) = \int_{-\infty}^x f(x) dx = \frac{x-A}{B-A} \text{ for } A \leq x \leq B$$

$$E(X) = \frac{A+B}{2} \quad \text{Var}(X) = \frac{(B-A)^2}{12}$$



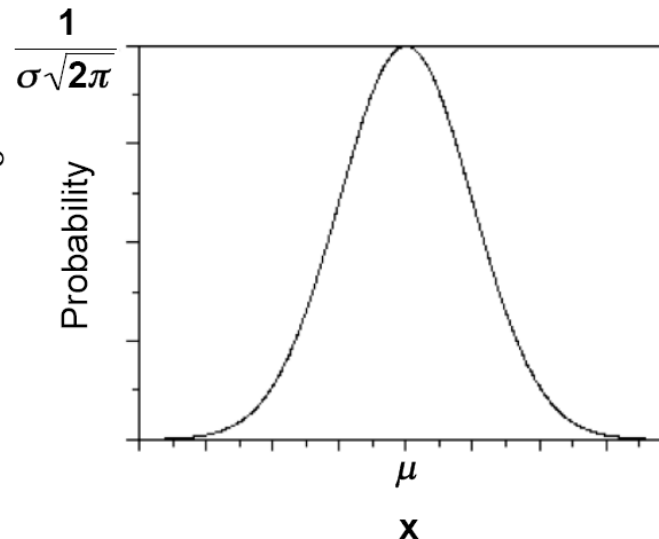
Normal distribution

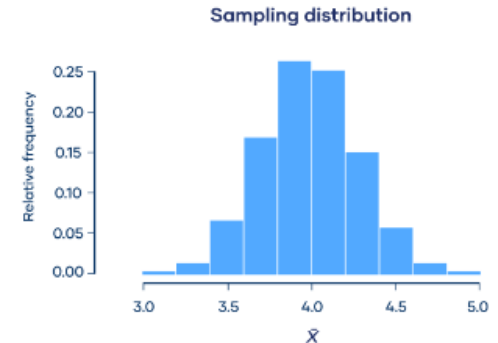
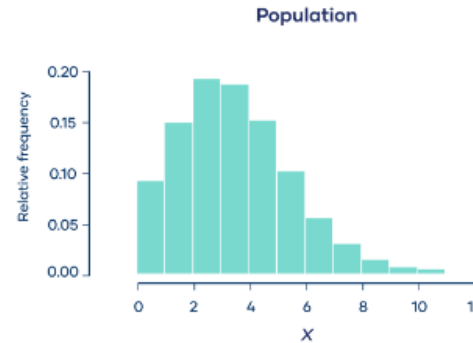
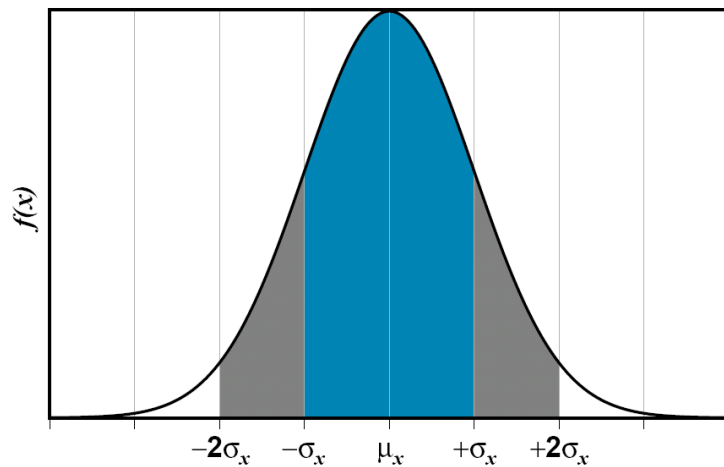
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \text{ for } -\infty \leq x \leq \infty$$

$$F(x) = \int_{-\infty}^x f(x) dx$$

must be evaluated numerically

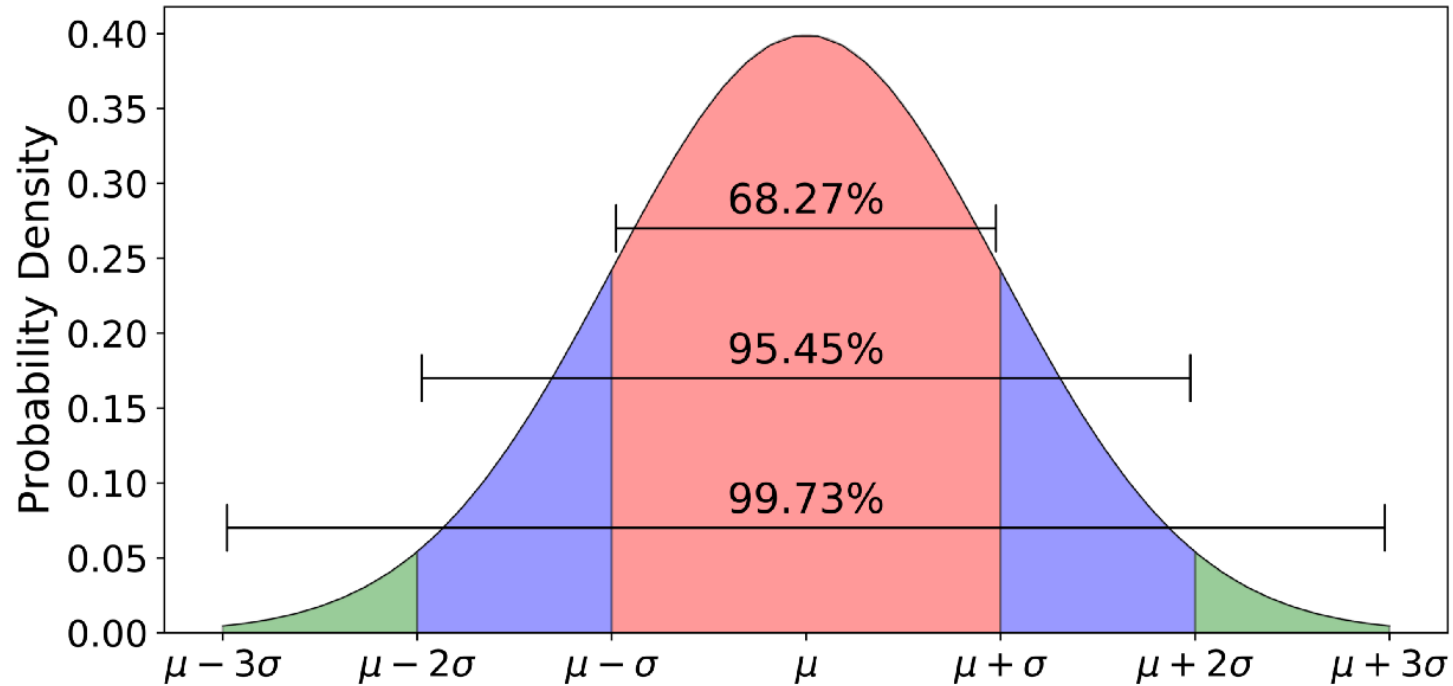
$$E(X) = \mu \quad \text{Var}(X) = \sigma^2$$





- ❖ Most commonly used distribution
- ❖ Many variables in nature are normally distributed
- ❖ Fully described by two parameters:
mean (μ_X) and standard deviation (σ_X)
- ❖ The mean = the median
- ❖ **Central Limit Theorem (CLT)**–
the sum of a large number of individual random components tends to be a normal distribution

68-95-99.7 Rule



Special Type of Normal Distribution

❖ Standardized Normal Distribution (표준정규분포)

- ◆ 정규분포 중 평균이 0, 표준편차가 1인 분포 $Z \sim N(0, 1^2)$
- ◆ 확률변수의 표준화를 거침 $X \sim N(\mu_X, \sigma_X^2)$

$$Z = (X - \mu_X) / \sigma_X$$

❖ Lognormal Distribution (대수정규분포)

- ◆ 만약 확률변수 X 와 Y 에 대해서 $Y = \ln(X)$ 가 성립하고,
- ◆ Y 가 정규분포를 따른다면 $Y \sim N(\mu_Y, \sigma_Y^2)$
- ◆ X 는 대수정규분포를 따른다고 할 수 있다 $X \sim LN(\mu_{\ln(X)}, \sigma_{\ln(X)}^2)$

The Standardized Normal Distribution

DEFINITION

The normal distribution with parameter values $\mu = 0$ and $\sigma = 1$ is called the **standard normal distribution**. A random variable having a standard normal distribution is called a **standard normal random variable** and will be denoted by Z . The pdf of Z is

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad -\infty < z < \infty$$

The graph of $f(z; 0, 1)$ is called the *standard normal* (or z) curve. Its inflection points are at 1 and -1 . The cdf of Z is $P(Z \leq z) = \int_{-\infty}^z f(y; 0, 1) dy$, which we will denote by $\Phi(z)$.

The Lognormal Distribution

DEFINITION

A nonnegative rv X is said to have a **lognormal distribution** if the rv $Y = \ln(X)$ has a normal distribution. The resulting pdf of a lognormal rv when $\ln(X)$ is normally distributed with parameters μ and σ is

$$f(x; \mu, \sigma) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} e^{-[\ln(x) - \mu]^2 / (2\sigma^2)} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Be careful here; the parameters μ and σ are not the mean and standard deviation of X but of $\ln(X)$. The mean and variance of X can be shown to be

$$E(X) = e^{\mu + \sigma^2/2} \quad V(X) = e^{2\mu + \sigma^2} \cdot (e^{\sigma^2} - 1)$$

Examples: Standardized Normal Distribution

EXAMPLE 4.16 The time that it takes a driver to react to the brake lights on a decelerating vehicle is critical in helping to avoid rear-end collisions. The article “Fast-Rise Brake Lamp as a Collision-Prevention Device” (*Ergonomics*, 1993: 391–395) suggests that reaction time for an in-traffic response to a brake signal from standard brake lights can be modeled with a normal distribution having mean value 1.25 sec and standard deviation of .46 sec. What is the probability that reaction time is between 1.00 sec and 1.75 sec? If we let X denote reaction time, then standardizing gives

$$1.00 \leq X \leq 1.75$$

if and only if

$$\frac{1.00 - 1.25}{.46} \leq \frac{X - 1.25}{.46} \leq \frac{1.75 - 1.25}{.46}$$

Thus

$$\begin{aligned} P(1.00 \leq X \leq 1.75) &= P\left(\frac{1.00 - 1.25}{.46} \leq Z \leq \frac{1.75 - 1.25}{.46}\right) \\ &= P(-.54 \leq Z \leq 1.09) = \Phi(1.09) - \Phi(-.54) \\ &= .8621 - .2946 = .5675 \end{aligned}$$

Exponential Distribution

❖ Exponential Distribution (지수 분포) $f_X(x) = \lambda e^{-\lambda x} \quad x \geq 0, \lambda > 0$

- ◆ 사건이 서로 독립적이고,
- ◆ 일정 시간동안 발생하는 사건의 횟수가 포아송 분포를 따를 때
- ◆ 다음 사건이 일어날 때까지의 대기 시간
- ◆ (=기하 분포, 그러나 연속적인 확률변수에 대해 적용이 가능)

4.2.2 Exponential distribution

From the Poisson process discussed in Subsection 4.1.3, the probability of no occurrences of the random variable X , which denotes a success, arrival, count, happening, or any other type of specified event, during a time interval t is

$$p_X(X = 0) = e^{-\lambda t}.$$

Using this result and considering the time T between occurrences as the random variable, we find the cdf of variable T is

$$F_T(T \leq t) = 1 - e^{-\lambda t}.$$

This means that the *waiting time* between successive events of a Poisson process has an *exponential distribution*. We note that, as in the Poisson, the distribution is applicable for other variables such as length and space in addition to time. Also, it follows that the exponential in continuous time corresponds to the geometric distribution in discrete time.

The pdf of the exponential distribution is written as follows by differentiating the expression just given with respect to t and by replacing T by a general variable X :

$$\begin{aligned} f_X(x) &= \lambda e^{-\lambda x}, & \text{for } x \geq 0, \lambda > 0, \\ &= 0, & \text{otherwise.} \end{aligned} \quad (4.2.3a)$$

For the same conditions, the cdf is

$$F_X(x) = 1 - e^{-\lambda x}. \quad (4.2.3b)$$

This is also referred to as the *negative exponential* because of the negative term in the exponent.

The Exponential Distribution

The family of exponential distributions provides probability models that are very widely used in engineering and science disciplines.

X is said to have an **exponential distribution** with (scale) parameter λ ($\lambda > 0$) if the pdf of X is

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (4.5)$$

Some sources write the exponential pdf in the form $(1/\beta)e^{-x/\beta}$, so that $\beta = 1/\lambda$. The expected value of an exponentially distributed random variable X is

$$\mu = E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

Obtaining this expected value necessitates doing an integration by parts. The variance of X can be computed using the fact that $V(X) = E(X^2) - [E(X)]^2$. The determination of $E(X^2)$ requires integrating by parts twice in succession. The results of these integrations are as follows:

$$\mu = \frac{1}{\lambda} \quad \sigma^2 = \frac{1}{\lambda^2}$$

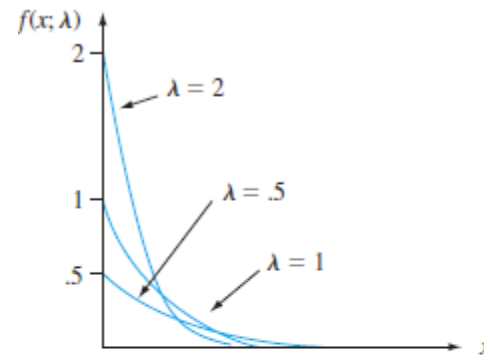


Figure 4.26 Exponential density curves

The exponential pdf is easily integrated to obtain the cdf.

$$F(x; \lambda) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Exponential Distribution Example

EXAMPLE 4.21 The article “Probabilistic Fatigue Evaluation of Riveted Railway Bridges” (*J. of Bridge Engr.*, 2008: 237–244) suggested the exponential distribution with mean value 6 MPa as a model for the distribution of stress range in certain bridge connections. Let’s assume that this is in fact the true model. Then $E(X) = 1/\lambda = 6$ implies that $\lambda = .1667$. The probability that stress range is at most 10 MPa is

$$P(X \leq 10) = F(10; .1667) = 1 - e^{-(.1667)(10)} = 1 - .189 = .811$$

The probability that stress range is between 5 and 10 MPa is

$$\begin{aligned} P(5 \leq X \leq 10) &= F(10; .1667) - F(5; .1667) = (1 - e^{-1.667}) - (1 - e^{-.8335}) \\ &= .246 \end{aligned}$$



Summary of Continuous Distributions

Table 4.2.2 Summary of continuous distributions

Distribution	Pdf	Mean, $E[X]$	Variance, $\text{Var}[X]$	Mgf
Uniform	$\frac{1}{b-a}$, for $a \leq x \leq b$; $-\infty < a < b < \infty$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt} - e^{at}}{t(b-a)}$
Exponential	$\lambda e^{-\lambda x}$, for $x \geq 0$; $\lambda > 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$, for $t < \lambda$
Gamma	$\frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}$, for $x \geq 0$; $\lambda, r > 0$	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$	$\left(\frac{\lambda}{\lambda - t}\right)^r$, for $t < \lambda$
Beta	$\frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$, for $0 < x < 1$; $\alpha, \beta > 0$	$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$	Not in a useful form
Weibull	$\frac{\beta}{\lambda} \left(\frac{x}{\lambda}\right)^{\beta-1} \exp\left[-\left(\frac{x}{\lambda}\right)^\beta\right]$, for $x, \beta, \lambda > 0$	$\lambda \Gamma\left(1 + \frac{1}{\beta}\right)$	$\lambda^2 \left\{ \Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2 \right\}$	$\lambda^t \left(1 + \frac{t}{\beta}\right)$
Normal	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$, for $-\infty < x, \mu < \infty$; $\sigma > 0$	μ	σ^2	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$
Lognormal ^a	$\frac{1}{x\sigma\sqrt{2\pi}} \exp\{-[\ln(x) - \mu]^2/2\sigma^2\}$, for $x, \sigma > 0$; $-\infty < \mu < \infty$	$\exp(\mu + \sigma^2/2)$	$\exp(2\mu + 2\sigma^2) - \exp(2\mu + \sigma^2)$	Does not exist

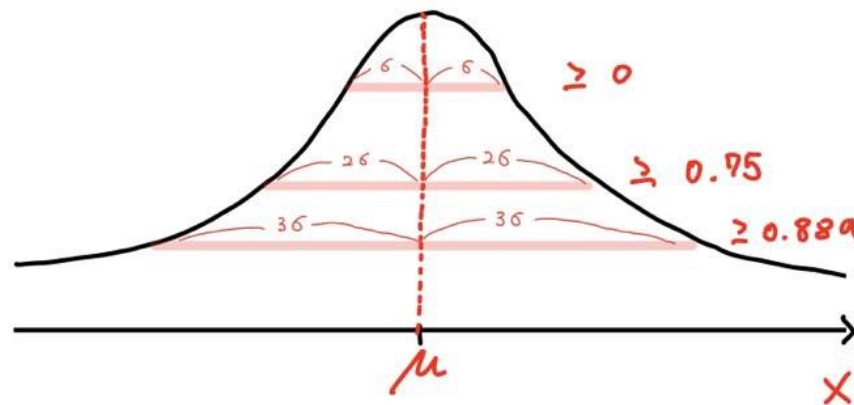
^a Note: μ and σ are the mean and standard deviations, respectively, of $\ln(x)$.

Chebyshev Inequality

$$P\{|X - \mu| \geq k\sigma\} \leq \frac{1}{k^2}$$

❖ Chebyshev Inequality (체비셰프 부등식)

- ◆ A single observation selected at random from any probability distribution will deviate more than $k\sigma$ from the mean μ , of the distribution with probability less than or equal to $1/k^2$.
- ◆ Similar to the 68-95-99.7 rule, but to ALL distributions
 - Provides a very conservative estimation of probability
- ◆ Assumes that the probability is **symmetric w.r.t. the mean μ** (if not, Cantelli Inequality, or one-tailed chebychev inequality)



Law of Large Numbers

- ❖ The (weak) Law of Large Numbers (큰 수의 약한 법칙)
 - ◆ 큰 모집단에서 무작위로 뽑은 표본의 평균이 전체 모집단의 평균과 가까울 가능성이 높다는 개념

Let $p_X(x)$ be a probability density function with mean μ_X and finite variance σ_X^2 . Let $\bar{x}_n$ be the mean of a random sample of size n from $p_X(x)$. Let ϵ and δ be any two specified small numbers such that ($\epsilon > 0, 0 < \delta < 1$). Then for n any integer greater than $\frac{\sigma_X^2}{\epsilon^2 \delta}$

$$\text{prob}(|\bar{x}_n - \mu_X| \geq \epsilon) \leq \delta \quad (3.79)$$

◆ Related Concepts

- Beginner's Luck (장기적으로는? 수렴하게 됨)
- 야구에서의 "DTD"

Conclusion of Lecture 2

❖ Few statistical characteristics common to many types of hydrologic data (수문 분야에서의 불확실한 변수들의 통계적 특성)

(Handbook of Hydrology, 17.1.1)

- ◆ Non-normal, skewed distribution functions, often having a lower bound of zero
- ◆ Outliers - a few values that are much larger or much smaller than the bulk of the data
- ◆ Serial correlation, or lack of independence among observations
- ◆ Distributions dependent on other variables
- ◆ Censoring – data concentration values reported as less than a detection limit, or discharge values less than the discharge at which flood damage occurs
- ◆ Seasonal patterns; statistical characteristics tend to vary with the seasons

Future Schedule Preview

- ❖ 다음 주 강의 (2022-09-20 화요일 18:30-21:20)
 - ◆ Lec 03: Probabilistic Modeling in Hydrology- Part II
(Maximum Likelihood Estimator & Goodness of fit tests)
 - Maximum Likelihood Estimation of Parameters & Examples
 - Goodness of fit Approaches such as:
AIC, PP or QQ plots, Chi-squared test
- ❖ Datacamp Assignment due 9/15 (Introduction to R)
- ❖ Assignment on Page 16 (Deriving the Poisson Distribution)
- ❖ **Bring LAPTOPS with R & Rstudio on Sep-27! (2주 후)**